a much more powerful technique had earlier been employed: radar. Radio waves effortlessly penetrate the clouds and thick atmosphere of Venus, bounce off the surface, and return to Earth, where they are gathered in and used to make a picture. The first work had been done, chiefly, by American ground-based radar at JPL's Goldstone tracking station in the Mojave Desert and at the Arecibo Observatory in Puerto Rico, operated by Cornell University.

Then the U.S. *Pioneer 12,* the Soviet *Venera 15* and *16* and the U.S. *Magellan* missions inserted radar telescopes into orbit around Venus and mapped the place pole to pole. Each spacecraft would transmit a radar signal to the surface and then catch it as it bounced back. From how reflective each patch of surface was and how long it took the signal to return (shorter from mountains, longer from valleys), a detailed map of the entire surface was slowly and painstakingly constructed.

The world so revealed turns out to be uniquely sculpted by lava flows (and, to a much lesser degree, by wind), as described in the next chapter. The clouds and atmosphere of Venus have now become transparent to us, and another world has been visited by the doughty robot explorers from Earth. Our experience with Venus is now being applied elsewhere—especially to Titan, where once again impenetrable clouds hide an enigmatic surface, and radar is beginning to give us hints of what might lie below.

VENUS HAD LONG BEEN THOUGHT of as our sister world. It is the nearest planet to the Earth. It has almost the same mass, size, density, and gravitational pull as the Earth does. It's a little closer to the Sun than the Earth, but its bright clouds reflect more sunlight back to space than our clouds do. As a first guess you might very well imagine that, under those unbroken clouds, Venus was rather like Earth. Early scientific speculation included fetid swamps crawling with monster amphibians, like the Earth in the Carboniferous Period; a world desert; a global petroleum sea; and a seltzer ocean dotted here and there with limestone-encrusted islands. While based on some scientific data, these "models" of Venus—the first dating